DAL DATA & AI LAB

UK Construction SME: AI Tender Worked Case, 2026

A synthetic, controlled adoption method

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GATE | EXTRACT | EVIDENCE | CHALLENGE | ASSURE

EXECUTIVE BRIEF

A synthetic tender workflow for controlled adoption

Northstar Build Ltd is a fictional 18-person contractor. The community-centre refurbishment, tender clauses, evidence and calculations are synthetic. The case demonstrates a method; it does not produce a real bid.

18 **fictional employees**	6 **workstations**	0 **real uploads**
Scenario input, not a sector average	Gate, extract, evidence, draft, challenge, assure	Browser-only deterministic experience

Case verdict

- AI may assist bounded requirement extraction, retrieval and first-draft structure.
- Human owners retain commercial, delivery, safety, regulatory and submission decisions.
- A pilot compares total effort and quality against the same baseline.
- Hold, revise and stop are valid outcomes.

01 THE FICTIONAL CASE

Northstar Build Ltd

Case item	Synthetic assumption	Status
People	18-person regional contractor	Invented scenario
Opportunity	Community-centre refurbishment	Invented tender
Problem	Repeated clause extraction and evidence assembly	Hypothesis to test
Inputs	Four fixed clauses and controlled evidence records	Synthetic browser data
Excluded	Real client data, prices, programmes, method statements and safety decisions	Hard control boundary
Decision	Human submission sign-off	Cannot be delegated

Why this workflow

Tender documents make requirements, evidence gaps and unsupported claims observable. The design can demonstrate safe stops without pretending to evaluate construction engineering or commercial judgement.

No upload, live model, client system or construction software connection is used.

02 SIX WORKSTATIONS

From opportunity gate to assurance

Stage	Action	Proceed signal
1 Opportunity gate	Check fit, capacity, deadline and prohibited assumptions	Named owner and conditional bid decision
2 Requirements	Extract and compare every clause with source wording	Mandatory requirements reviewed
3 Evidence	Match current, relevant and owned evidence	Gaps remain visible; evidence approved
4 Draft control	Prepare a source-linked response skeleton	Every material statement has a source and reviewer
5 Challenge	Test delivery, commercial and commitment assumptions	Issues resolved, assigned or held
6 Assurance	Run final human compliance and quality review	Accountable person authorises any real submission

The simulation shows READY FOR HUMAN SIGN-OFF only after all synthetic review, evidence and challenge controls are complete. It never submits anything.

03 PILOT GATES

Proceed, revise, hold or stop

Gate	Proceed only when	Stop or revise when
G0 Scope	Bid decision, owner and prohibited actions named	Generic objective or unclear accountability
G1 Data	Approved documents, access and retention defined	Sensitive data enters an unapproved tool
G2 Known cases	Extraction and retrieval pass fixed tests	Material omission, invention or broken citation
G3 Shadow	Current process remains authoritative	Users cannot explain or override output
G4 Pilot	Quality, review and total cost stay acceptable	Draft speed disappears after correction
G5 Scale	Monitoring, support and rollback are approved	Unresolved delivery, safety, commercial or lock-in risk

Zero-tolerance boundaries

Unauthorised disclosure, invented mandatory requirements, unapproved commitments and autonomous safety or regulatory decisions trigger a stop or escalation.

04 MEASUREMENT AND REUSE

Replace every assumption with a local baseline

Measure	Record
Baseline	Preparation, review, correction, quality and incident measures for the current workflow
Pilot	The same measures after assistive tooling and controls
Full cost	Licences, setup, training, review, correction, assurance and incidents
Capacity signal	Baseline hours minus pilot preparation and added review/correction
Decision	Scale, limit, revise, hold or stop with reasons

A positive hour balance is not ROI. Quality, control, full cost and opportunity cost remain part of the decision.

Sources

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2. RICS responsible use of AI standard
https://www.rics.org/profession-standards/rics-standards-and-guidance/conduct-competence/responsible-use-of-ai

3. NCSC construction information-security guidance
https://www.ncsc.gov.uk/news/joint-ventures-construction-guidance

4. Construction bid-preparation research prototype
https://www.tandfonline.com/doi/abs/10.1080/15623599.2026.2629023

Boundary

Evidence-informed fictional composite. Not construction, legal, safety, regulatory, procurement or investment advice.